

Viacheslav Klimenko

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Appointments	Postdoctoral Fellow	Oct. 2022 – Present
	Department of Physics and Astronomy, University of South Carolina, Columbia, SC, USA	
	Postdoctoral Fellow	Apr. 2022 – Sep. 2022
	Department of Physics and Astronomy, University of South Carolina, Columbia, SC, USA	
	Research Scientist	2016 – Apr. 2022
	Department of Applied Mathematics and Mathematical Physics, Ioffe Institute, St. Petersburg, Russia	
Education	Assistant Professor	2016 – 2021
	Department of Astronomy, St. Petersburg Academic University, Russian Academy of Sciences, St. Petersburg, Russia	
	Junior Research Scientist	2011 – 2016
	Department of Theoretical Astrophysics, Ioffe Institute, St. Petersburg, Russia	
Education	Ph.D. in Astronomy & Astrophysics	2016
	Ioffe Institute, St. Petersburg, Russia	
	M.Sc. in Theoretical Physics	2011
	St. Petersburg Academic University, St. Petersburg, Russia	
Education	B.Sc. in Physics	2009
	Peter the Great St. Petersburg Polytechnic University, St. Petersburg, Russia	
Summary	Projects Funding: Total of \$2.3M in external awards as PI/Co-I/Collaborator (~\$180K as PI and Co-I at USC).	
	Publications: 25 refereed manuscripts published in or submitted to peer-reviewed journals	
	- 14 first-author peer-reviewed manuscripts (<i>ApJ</i> , <i>MNRAS</i>).	
	- 2 best scientific work of the year (<i>Astronomy Letters Journal</i> , <i>Ioffe Institute</i>).	
	- Invited to be a referee for <i>Monthly Notices of the Royal Astronomical Society</i> , <i>Astronomy & Astrophysics</i> , <i>Astrophysical Journal</i> , <i>Physics Conference Series</i> .	
	Project management: Lead and participate in several international projects	
	- probing the gas in and around local galaxies mapped with Integral Field Spectroscopy	
	- evolution of dust composition and dust extinction across cosmic time	
	- comprehensive view of molecular gas in distant galaxies	
	- isotopic abundance ratios and fundamental physics at cosmic noon	
Summary	Observing Experience: PI and Co-I of 13 approved proposals using 2–10m class telescopes from ultraviolet to mid-infrared.	
	Mentoring: 3 graduate students, and 4 undergraduate students.	
	Teaching: Taught 2 undergraduate-level courses as Assistant Professor and 2 as Instructor.	
	Public Outreach: Public lectures at the St. Petersburg Planetarium; <i>Astronomy on Tap</i> ; STEM Hero Fair at USC; Midway Physics Day at the South Carolina State Fair; and Discover USC.	
	Diversity and Inclusion: I am a first-generation scholar from a low-income, nontraditional background. I pursued a path in academic research, established an independent research program, and built collaborations with researchers around the globe from diverse cultural, national, and educational backgrounds. I also value a balanced life that integrates research, family, and outdoor activities (e.g., mountaineering, climbing, running).	

Teaching Experience	Guest Lecturer University of South Carolina, Columbia, SC – ASTR201: Introduction to Astronomy: The Dark Universe - <i>Spring</i> 2025 – ASTR 301: Galactic Astrophysics- <i>Spring</i> 2026	
	Assistant Professor Department of Astronomy, St. Petersburg Academic University, St. Petersburg, Russia Statistical Mechanics - <i>Fall, Spring</i> 2016–2021 Introduction to Modern Astronomy - <i>Spring</i> 2019	
	Teaching Specialist Peter the Great St. Petersburg Polytechnic University, St. Petersburg, Russia General physics - <i>Fall, Spring</i> 2011–2014	
Mentoring	Graduate students: Ioffe Institute - Kurichin, Oleg (2019-2022, comentoring with A. Ivanchik) University of South Carolina - Putha, Karthik (2023-present, comentoring with V. Kulkarni) - Santhigiri, Ananthavishnu (2024-present, comentoring with V. Kulkarni)	
	Undergraduate students: University of South Carolina - Lee, Jack (Spring 2024-Fall 2025) - David, Hoffman (Fall 2025-present, comentoring with V. Kulkarni) St. Petersburg Academic University, - Shenyavsky, Vsevolod (2020-2022) Peter the Great St. Petersburg Polytechnic University, - Baryshnikov, Ilya (2020-2022)	
Research Funding	As a Principal Investigator:	
	NASA – Keck Principal Investigator Data Award, 2026	\$15K
	Russian Foundation for Basic Research – Interstellar Ultraviolet Radiation in Distant Galaxies, 2018–2019	\$15K
	Dynasty Foundation – Physical Conditions and Chemistry of the ISM/IGM at High Redshift, 2014	\$10K
	As a Co-investigator/Collaborator	
	JWST Cycle 3 – Mining for DIAMONDS (Dust, Ice, And MOlecules iN Distant Galaxies) over the Past ~7 Billion Years (Co-I), 2024–2027	\$244K
	JWST Cycle 3 – Probing the Dust in the Diffuse ISM in Galaxies over the Past ~8 Billion Years with Spectroscopy of Background Quasars (Co-I), 2024–2027	\$264K
	NASA – Mapping the Baryon Cycle In and Around Galaxies Through Cosmic Time (collaborator), 2025–2028	\$558K
	NASA – Characterizing Interstellar Dust Extinction in Galaxies and Its Cosmic Evolution (collaborator), 2024–2027	\$397K
	Russian Science Foundation – Physics of the Early Universe: Observational Cosmology (Co-I), 2018–2022	\$400K
	Russian Foundation for Basic Research – Molecular and Diffuse ISM at High Redshifts (Co-I), 2018–2019	\$29K
	Russian Science Foundation – Physics of the Early Universe: Chemistry and Physical Conditions (Co-I), 2014–2016	\$297K
	Russian Foundation President Programme – Galaxies in the Early Universe (Co-I), 2013–2014	\$32K
	Russian Foundation for Basic Research – Interstellar Molecular Hydrogen Clouds at Early Stages (Co-I), 2012–2013	\$21K
	Russian Foundation for Basic Research – Chemistry of the ISM at High Redshift (Co-I), 2011–2012	\$61K

Telescope Time	As a Principal Investigator:	
	VLT/X-shooter – “SUENO: Survey of Unexplored Extinction in Normal galaxies with X-shooter Observations”, PI: V. Klimenko	26.6h, 2026
	Keck/NIRES - “The Keck PRISM Survey: Probing Reddening in Intervening Silicate Dust with Near-Infrared Spectroscopy”, PI: V. Klimenko	1 night, 2026
	SAO/SCORPIO - “Observing the emission of ionized gas in galaxies with a strong halo”, PI: V. Klimenko	25h, 2025
	As a Co-investigator	
	Gemini/GNIRS – “Unveiling the Nature of DLA Host Galaxies: NIR Spectroscopy of Nebular Emission at $z \sim 2$ ”, PI: A. Shaban, (Co-I)	20.4h, 2026
	HST/STIS – “Multiwavelength Extinction Curves in Distant Galaxies”, PI: M. Aller, (Co-I)	26 orbits, 2026
	GEMINI/FLAMINGOS2 - “Galactic superwinds and their role in shaping the ultra-strong metal absorbers”, (Co-I), PI: Varsha Kulkarni	20h, 2025
	JWST/MIRI – Cycle 3 - “Mining for DIAMONDS (Dust, Ice, And MOlecules in Distant Galaxies) over the Past ~ 7 Billion Years” (Co-I), PI: V. Kulkarni	25h, 2024
	JWST/MIRI – Cycle 3 - “Probing the Dust in the Diffuse ISM in Galaxies over the Past ~ 8 Billion Years with Spectroscopy of Background Quasars” (Co-I), PI: V. Kulkarni	25h, 2024
	VLT/ESPRESSO – “Uncovering the Fossil Record of Chemical Evolution with Isotopic Abundance Ratios at Cosmic Noon” (Co-I), PI: V. Kulkarni	18h, 2023
	VLT/ESPRESSO – “ESPRESSO Observations of a Unique CO Absorber: Paving the Way to Fundamental Physics and Cosmology with ELT-ANDES” (Co-PI), PI: S. Lopez	18h, 2022
	LBT/PEPSI – “Neutral Gas in and around Nearby Galaxies Mapped by MaNGA” (Co-I), PI: J. Hill	10h, 2022
	VLT/X-shooter – “Probing the Cold and Dense Interstellar Medium at High Redshifts Using Saturated H ₂ Absorption Systems” (Co-I), PI: S.A. Balashev	16h, 2015
	VLT/UVES – “Molecular Hydrogen Absorption Systems at High Redshifts: Physical Conditions in the Interstellar Medium at the Early Universe” (Co-I), PI: D.A. Varshalovich	9h, 2014

Research Collaborations

University of South Carolina (USA):
 Varsha Kulkarni (Dust and Gas in Distant Galaxies)
 Mehrnoosh Tahani (Galactic Magnetic Fields, Dragonfly Polarimetry)

University of Wisconsin-Madison (USA):
 Matthew Bershadsky (Integral Field Spectroscopy of Galaxies)

Georgia State University (USA):
 Monique Aller (Dust in distant Galaxies)

University of North Carolina Asheville (USA):
 David Wake (Integral Field Spectroscopy of Galaxies)

Space Telescope Science Institute (USA):
 Karl Gordon (Galactic Dust Extinction)

Presbyterian College (USA):
 Frances H. Cashman (High Galactic latitude gas and dust.)

Macquarie University (Australia):
 Tayyaba Zafar (Extragalactic dust extinction.)

European Southern Observatory (Germany/Chile):
 Pasquier Noterdaeme (Molecular absorbers)
 Celine Peroux (Integral Field Spectroscopy of Galaxies)
 Annalisa DeCia (Isotopic abundances, Dust depletion and extinction)

Ioffe Institute (Russia):
 Sergey Balashev (Molecular absorption systems, ISM gas models)

Inter-University Centre for Astronomy & Astrophysics (India):
 Raghunathan Srianand (Molecular absorption systems)

Software	Developer of publicly available Python packages for dust-extinction modeling, ISM physical-condition analysis, custom procedures for HST COS/STIS, JWST MIRI MRS pipelines (github.com/slavaklimenko).	
Awards	Best scientific work of the year, <i>Astronomy Letters Journal</i>	2020
	Best scientific work of the year, <i>Ioffe Institute</i>	2017
	Invited talks at conferences and institutions	Several
Publication List	Submitted: 1. Klimenko, V. V., Aller, M. C., & Kulkarni, V. P., "Mining for Diamonds in a Lensed Galaxy ~ 7 Billion Years Ago," <i>submitted to Nature</i>, 2026. Published: 1. Klimenko V.V., Aller M.C., & Kulkarni V.P., "Probing Dust Composition in Distant Galaxies with JWST Mid-IR Spectroscopy of Quasars with Foreground 2175 Å Absorbers II. Measurements of Grain Composition and Extinction Properties," <i>ApJ</i>, 2026, 1003, 129. 2. Klimenko V.V., Aller M.C., & Kulkarni V.P., "Probing Dust Composition in Distant Galaxies with JWST Mid-IR Spectroscopy of Quasars with Foreground 2175 Å Absorbers I: Methodology," <i>ApJ</i>, 2026, 998, 299. 3. Baryshnikov I.A., Shenyavskiy V.A., Klimenko V.V., Ivanchik A.V., "Modification of the method for measuring the CMB temperature based on the Sunyaev–Zeldovich effect," <i>Technical Physics</i>, 2024, 69, 1922. 4. Klimenko V.V., Balashev S.A., Noterdaeme P., Srianand A., "Excitation of CO molecules in diffuse gas over cosmic history," <i>MNRAS</i>, 2024, 533, 1367. 5. Kosenko D.N., Balashev S.A., Klimenko V.V., "Cold diffuse interstellar medium of Magellanic Clouds: II. Physical conditions from excitation of C I and H₂," <i>MNRAS</i>, 2024, 528, 5065. 6. Klimenko V.V., Kulkarni V.P., Wake D.A., Poudel S., Bershadsky M.A., Peroux C., Lundgren B., "The Baryonic Content of Galaxies Mapped by MaNGA and the Gas Around Them," <i>ApJ</i>, 2023, 954, 115. 7. Kurichin O.A., Kislitsyn P.A., Klimenko V.V., Balashev S.A., Ivanchik A.V., "A new determination of the primordial helium abundance using the analyses of H II region spectra from SDSS," <i>MNRAS</i>, 2021, 502, 3045. 8. Klimenko V.V., Balashev S.A., "Physical conditions in the diffuse interstellar medium of local and high-redshift galaxies: measurements based on the excitation of H₂ rotational and C I fine-structure levels," <i>MNRAS</i>, 2020, 498, 1531. 9. Klimenko V.V., Ivanchik A.V., Petitjean P., Noterdaeme P., Srianand R., "Estimation of the Cosmic Microwave Background Temperature from Atomic C I and Molecular CO Lines in the Interstellar Medium of Early Galaxies," <i>Astronomy Letters</i>, 2020, 46, 715. 10. Klimenko V.V., Ivanchik A.V., "The measurements of the CMB temperature in diffuse interstellar medium of the Milky Way and high-redshift galaxies based on excitation of C I fine-structure and H₂ rotational levels," <i>Journal of Physics: Conference Series</i>, 2020, 1697, 012013. 11. Klimenko V.V., Petitjean P., Ivanchik A.V., "Observational estimate of the partial covering probability of quasar emission regions by distant H₂ absorption clouds," <i>MNRAS</i>, 2020, 493, 5743. 12. Klimenko V.V., Ivanchik A.V., "Influence of Radiative Pumping on the HD Rotational Level Populations in Diffuse Molecular Clouds of the Interstellar Medium," <i>Astronomy Letters</i>, 2020, 46, 224. 13. Balashev S.A., Klimenko V.V., Noterdaeme P., Krogager J.-K., Varshalovich D.A., Ivanchik A.V., Petitjean P., Srianand R., Ledoux C., "X-shooter observations of strong H₂-bearing DLAs at high redshift," <i>MNRAS</i>, 2019, 490, 2668.	

14. Balashev S.A., **Klimenko V.V.**, Noterdaeme P., Krogager J.-K., Ledoux C., Ivanchik A.V., Varshalovich D.A., Petitjean P., "The cold neutral phase of the interstellar medium in high-redshift galaxies," *Journal of Physics: Conference Series*, 2019, 1400, 022030.
15. Kurichin O.A., Ivanchik A.V., **Klimenko V.V.**, "Primordial ^4He abundance refinement using sample of SDSS DR15 galaxies," *Journal of Physics: Conference Series*, 2019, 1400, 022051.
16. Balashev S.A., Noterdaeme P., Rahmani H., **Klimenko V.V.**, Ledoux C., Petitjean P., Srianand R., Ivanchik A.V., Varshalovich D.A., "CO-dark molecular gas at high redshift: very large H_2 content and high pressure in a low-metallicity damped Lyman alpha system," *MNRAS*, 2017, 470, 2890.
17. **Klimenko V.V.**, Balashev S.A., Ivanchik A.V., Varshalovich D.A., "Estimation of physical conditions in the cold phase of the interstellar medium in the sub-DLA system at $z = 2.06$ in the spectrum of the quasar J 2123–0050," *Astronomy Letters*, 2016, 42, 137.
18. **Klimenko V.V.**, Balashev S.A., Ivanchik A.V., Varshalovich D.A., "A new estimation of $\text{HD}/2\text{H}_2$ at high redshift using the spectrum of the quasar J 2123–0050," *Journal of Physics: Conference Series*, 2015, 661, 012005.
19. Ofengeim D.D., Balashev S.A., Ivanchik A.V., Kaminker A.D., **Klimenko V.V.**, "Effect of a partial coverage of quasar broad-line regions by intervening cloud," *Astrophysics and Space Science*, 2015, 359, 26.
20. **Klimenko V.V.**, Balashev S.A., Ivanchik A.V., Ledoux C., Noterdaeme P., Petitjean P., Srianand R., Varshalovich D.A., "Partial covering of the emission regions of Q 0528-250 by intervening H_2 cloud," *MNRAS*, 2015, 448, 280.
21. Balashev S.A., Noterdaeme P., **Klimenko V.V.**, Petitjean P., Srianand R., Ledoux C., Ivanchik A.V., Varshalovich D.A., "Neutral chlorine and molecular hydrogen at high redshift," *Astronomy & Astrophysics*, 2015, 575, L8.
22. Ivanchik A.V., Balashev S.A., Varshalovich D.A., **Klimenko V.V.**, " H_2/HD molecular clouds in the early universe. An independent means of estimating the baryon density of the Universe," *Astronomy Reports*, 2015, 59, 100.
23. Balashev S.A., **Klimenko V.V.**, Ivanchik A.V., Varshalovich D.A., Petitjean P., Noterdaeme P., "Molecular hydrogen absorption systems in Sloan Digital Sky Survey," *MNRAS*, 2014, 440, 225.
24. **Klimenko V.V.**, Ivanchik A.V., Varshalovich D.A., Pavlov A.G., "Influence of gamma-ray emission on the isotopic composition of clouds in the interstellar medium," *Astronomy Letters*, 2012, 38, 364.

Ongoing Publications

1. Putha K., **Klimenko V.V.**, Kulkarni V.P., "Constraints on the Chemical Composition of $z > 6$ galaxies and Population III Star Properties from JWST Quasar Spectroscopy," *ApJ*, *in prep*, 2026
2. Shaban A., **Klimenko V.V.**, Kulkarni V.P., "Survey of DLA Host Galaxies at $z \sim 2-3$ in the KCWI Archive," *A&A in prep.*, 2026
3. **Klimenko V.V.**, Kulkarni V.P., Wake D.A., Bershadsky M.A., Peroux C., Lundgren B., "The COS-MaNGA project. Analysis of the CGM around SDSS MANGA galaxies."

Talks and Conferences

- 11/2025 – Tucson, AZ: Poster at Dusty Universe 2025: “Extinction Curves from UV to Mid-IR from Dust In and Around Distant Galaxies”
- 06/2025 – Princeton University, NJ, USA: “Probing Dust Composition in Distant Galaxies with JWST Mid-IR Spectroscopy of Quasars with Foreground 2175Å Absorbers”
- 04/2025 – Francis Marion University, SC, USA: “The Baryonic Content of Galaxies Mapped by MaNGA and the Gas Around Them”
- 11/2023 – Santiago, Chile: “Cool-gas metallicity distribution around nearby galaxies mapped with integral field spectroscopy”
- 12/2019 – IKI RAS, Moscow, Russia: Poster: “Radiative pumping of HD rotational levels in the diffuse ISM”
- 12/2019 – SAO RAS, Russia: Poster: “Interstellar UV radiation field in high redshift galaxies probed by Damped Lyman Alpha systems”
- 04/2019 – Pushino, Russia: “Physical conditions in the diffuse ISM probed by excitation of H₂ rotational levels”
- 12/2018 – IKI RAS, Moscow, Russia: “Physical conditions in translucent clouds at high redshift probed by excitation of CO rotational levels”
- 12/2017 – IKI RAS, Moscow, Russia: “Measurements of the CMB temperature in the early Universe”
- 04/2015 – Pushino, Russia: “Physical conditions of the cold phase of the neutral ISM at high redshift”
- 12/2014 – IKI RAS, Moscow, Russia: “Physical conditions in dense molecular clouds probed by the analysis of excitation of H₂/C I levels in quasar spectra”
- 12/2014 – Swinburne University, Kuching, Malaysia: Poster: “HD/H₂ molecular clouds at high redshifts”
- 06/2014 – St. Petersburg, Russia: “The effect of partial covering of QSO emission regions by distant absorption clouds”
- 05/2013 – St. Petersburg, Russia: “Measurements of the variation of electron-to-proton mass ratio from the analysis of H₂ absorption systems in quasar spectra”
- 05/2011 – Catania, Italy: “The contribution of longitudinal plasmons to the thermodynamics of dense electron plasma”
- 10/2010 – St. Petersburg, Russia: “Influence of the gamma-ray emission on the isotopic composition of clouds in the interstellar medium”